MILITARY
SPECIFICATION

MIL-T-11157B(EL)
6 September 1963
Superseding
MIL-T-11157A(SigC)
27 October 1953

THEODOLITE, DOUBLE CENTER ML-474( )/GM

1. SCOPE

1.1 This specification covers one type of meteorological equipment used for tracking meteorological balloons to determine upper air wind velocity and direction, consisting of the following units and designated as Theodolite, Double Center ML-474( )/GM. (See 6.3.)

Main Telescope
Telescope ML-146( )
Compass ML-197( )
Case CY-787( )/U

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Flat, Steel.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Locked Corner.
PPP-636	Boxes, Fiber, Sheet and Cut Shape.
PPP-F-320	Fiberboard.
PPP-T-76	Tape, Pressure Sensitive Adhesive, Paper Water Resistant.

MILITARY

MIL-B-18	Batteries, Dry.
MIL-W-76	Wire and Cable, Hook-up Electrical, Insulated.
MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags, Interior Packaging.
MIL-B-131	Barrier-Material, Water Vaporproof, Flexible.
MIL-D-3464	Desiccants, Activated (in bags) for Static Dehumidification and Packaging.
MIL-M-13231	Marking of Electronic Items.

FSC 6660

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MIL-P-11268	Parts, Materials and Processes used in Electronic Communication Equipment.
MIL-F-14072	Finishes for Ground Signal Equipment.
JAN-F-675	Films, Reflection Reducing, for Glass Optical Elements.

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking of Shipments.
MIL-STD-169	Extreme Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle.
MS 20003	Indicator, Humidity, Card, Three Spot Impregnated Areas (Cobaltus Chloride).

DRAWINGS

ELECTRONICS COMMAND

SC-DL-9073	Case CY-787( )/U.
SC-A-46439	List of Accessories for Packaging.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Preproduction samples.- The contractor shall furnish two preproduction samples for approval if required by the invitation for bid or request for proposal and contract. (See 4.3.)

3.2 Procurement model.- A procurement model of the theodolite will be available for inspection by bidders and will be loaned to the contractor. Unless otherwise specified herein or in the invitation for bids or request for proposal, physical construction of the theodolite shall conform to the model. (NOTE: In case of conflict between specified performance characteristics for the theodolite and the performance of the model, the specified performance characteristics shall govern.)

3.3 Parts, materials and processes.- In addition to the requirements of this specification, the requirements of Specification MIL-P-11268, including the selection requirements therein, shall apply.

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3.4 Finish.- The equipment, including the case, shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on Type I surfaces shall be green (olive drab) color, semigloss enamel matching a color chip provided by the procuring agency. (See 6.2.)

3.5 Detail and operational requirements.- These requirements are as specified in 3.6 through 3.10.

3.6 Main telescope.- The main telescope shall meet the following mechanical and optical requirements.

3.6.1 Assembly.- The telescope assembly shall be balanced by counterweights in such a manner that it will remain in any position in which it is placed without tendency to rotate either vertically or horizontally when all tangent screws or locking devices are released.

3.6.2 Prism adjustment.- Adjustment of the prism shall be accomplished by means of oppositely directed screws bearing on projecting surfaces of the prism holder; these surfaces shall be flat and parallel. The ends of the adjusting screws bearing thereon shall be firmly attached to the telescope housing so that loosening or changes in alignment will not occur under severe vibration. All openings in the telescope housing to permit parts of the prism holder to project outside the housing shall be provided with moisture-and dust-resistant covers.

3.6.3 Focusing.- It shall be possible to focus the telescope for all distances greater than 100 feet. The focusing mechanism shall move freely throughout its traverse, but there shall be no noticeable looseness in the fit of the movable focusing element in the telescope housing. Accidential change in the position of the focusing mechanism as the telescope is rotated shall be impossible. The difference in error of collimation between that obtained by focusing on a point of sufficient distance to be equivalent to infinite focus and that by focusing on a point at minimum focal distance shall not exceed an angle of 20 seconds of arc.

3.6.4 Eyepiece.- The eyepiece shall be of the achromatic type. It shall focus on the cross hairs in the telescope by means of a spiral movement. This movement shall correspond to a range of ±5 diopters from normal. The mechanical axis of the eyepiece shall coincide with the horizontal axis of rotation of the telescope. The lenses of the optical system shall be carefully centered to coincide with the geometrical axes of the instrument. No interference to the adjustment of the eyepiece mounting and focusing tube shall result from inserting the swinging mirror of the finder telescope.

3.6.5 Sights.- The accuracy of the sights shall be such that when the telescope is tested in accordance with 4.22 the image of a sighted object shall lie within the field of the telescope.

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3.6.6 Lenses.- The lenses in the telescope shall be of high grade and completely free from air bubbles at cemented surfaces. It shall be possible to focus a recognizable image of the object throughout the entire field of the telescope. The definition of the image shall be sharp, the illumination uniform, and the field sensibly flat at the center of the field and throughout an area equivalent to not less than 3/4 degree of field in all directions from this center. The image shall be sensibly free from distortion, coma, and spherical aberration. Chromatic aberration shall be sufficiently small so that the sharpness of the image is not affected. All air to glass surfaces except the outside of the objective and the eyepiece shall be given a non-reflective coating conforming to Specification JAN-F-675. The lenses shall be mounted in a manner that will produce a minimum strain due to differences in the thermal coefficient of expansion of the lens and its mounting.

3.6.7 Optical characteristics.- The optical characteristics of the telescope shall be as follows:

3.6.8 The magnifying power shall be not less than 19 diameters nor more than 24 diameters.

3.6.9 The field of view shall be not less than 40 degrees divided by the magnifying power.

3.6.10 The effective aperture of the objective shall be not less than 40 mm.

3.6.11 The eye distance shall not be less than 10 mm. nor more than 15 mm.

3.7 Finder telescope ML-146( ).- This telescope shall possess the following mechanical and optical characteristics.

3.7.1 Lever.- It shall be possible to move the swinging mirror to an in or out position by means of the lever without displacing the azimuth or vertical settings (gears engaged) of the main telescope by more than 40 seconds of arc. The mirror shall intercept the optical axis of the eyepiece when the lever is in a position toward the vertical scale of the theodolite.

3.7.2 Magnification.- The finder telescope shall possess a magnifying power of between 3.5 and 5 diameters and a field of view of not less than 10 degrees.

3.7.3 Focus.- The finder telescope shall be of a fixed focus type, focused for objects beyond 500 feet distance, and shall be adjusted so that when the image of a target at least 500 feet distant is accurately placed on the cross hairs, using the finder objective, the image shall be not more than one eighth of the field diameter from the cross hair intersection as determined by sighting the main objective on the same target without otherwise shifting the line of sight of the theodolite.

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3.7.4 Exit pupils and eye distances.- The exit pupil when using the finder objective shall be not less than 10 millimeters nor more than 20 millimeters.

3.8 Compass ML-197( ).- This compass shall have the following characteristics.

3.8.1 Balances.- The compass needle balances shall be of non-magnetic wire having sufficient tension around the needle so that after being once adjusted to balance, the needle will not slip due to jarring from any cause.

3.8.2 Lever and cam.- It shall be possible to lock the compass needle by means of a lever-actuated cam. The cam, lever and needle shall not become unlocked during normal use, due to vibration.

3.8.3 Adjustments.- The needle lock shall turn easily and without binding. The locking spring shall not damage the jewel bearing when the compass is locked. The arc of the cam shall not compress the spring to such an extent that unnecessary wear is apparent when the compass is unlocked.

3.8.4 Needle. The needle shall be supported on a jewel bearing. The center of gravity of the needle shall be at or near the point of support to reduce "needle quiver" to a minimum. The two ends of the needle and the jewel bearing shall be in the same plane.

3.9 Mechanical and physical characteristics, theodolite.- The mechanical and physical characteristics of the theodolite shall be as follows:

3.9.1 Levels.- The glass vials shall be ground with uniform curvature. The level sensitivity shall be not less than 100 seconds per 2 millimeters of bubble movement. The hysteresis shall be less than 20 seconds. Both level bubbles shall move along the vials without noticeably abrupt changes in sensitivity.

3.9.2 Intermediate center and leveling head.- The theodolite shall be provided with an intermediate center so that azimuth measurements can be made with reference to an established azimuth line. An azimuth scale shall be engraved on a circular plate attached to the intermediate center. The edge of this plate (scale) shall be concentric with the axis of the intermediate center to within .001 inch.

3.9.3 Looseness of centers.- The inner center shall be fitted in the inside cone bearing of the intermediate center, and the outside cone bearing of the intermediate center shall be fitted in the leveling-head. These bearing surfaces shall be hand-fitted so that the weight on the bearings is divided between the conical surfaces of the centers and horizontal ring, or shoulder, bearing surfaces around the upper edge of the centers. The fit shall be such that the greater part of the weight is carried by the shoulders. However, the conical surfaces shall fit so that looseness in the centers will not exceed 40 seconds of arc. Each center shall turn freely in each direction without drag. The leveling-head and intermediate center shall fit with at least .001 inch clearance between their ends and the bottom nut on the

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end of the inner center. The centers shall be slightly relieved in the center portion between bearing surfaces. They shall be fitted to provide bearing contact for approximately 1/3 of their length at each end.

3.9.4 Obliquity.- The axis of the inner conical surface of the intermediate center shall be in alignment with the axis of the outer conical surface so that the error of obliquity, shall not exceed 20 seconds of arc.

3.9.5 Tangent mechanisms.- The spring tension on the mechanisms in the in-mesh position shall be sufficient to meet the test requirements of 4.17.1. An accurate worm gear shall be cut on the edge of the vertical circle and on the edge of the intermediate-center plate. These treads shall be cut with 360 teeth about the circumference of each circle so that one complete revolution of the tangent screw will rotate the telescope one degree. The threads on the tangent screws shall fit the circle gears with sufficient accuracy to meet the test requirements of 4.17.3. The tangent mechanisms shall be aligned so that the screws may be placed in the in-mesh position without producing any noticable rotation of the circle plates. Total backlash shall not exceed an angular displacement of 0.02 degree.

3.9.6 Adjustment of the theodolite.- Each theodolite shall be capable of being adjusted for minimum error in the adjustments listed in 3.9.6.1 through 3.9.6.4 inclusive. The errors shall not exceed the tolerances given in 3.9.6.1 to 3.9.6.4 inclusive.

3.9.6.1 Adjustment of the vertical axis.- It shall be possible to adjust the vertical axis of the theodolite by means of the level adjustments inherent in the instrument so that there will be no noticeable error other than that which may be attributed to looseness of centers.

3.9.6.2 Adjustment of the line of collimation.- It shall be possible to adjust the line of collimation of the main telescope so that, with the vertical circle reading zero, it is horizontal when the instrument is leveled and so that the part of it in front of the prism is perpendicular to the horizontal axis of the telescope.

3.9.6.2.1 Accuracy. It shall be possible to adjust the cross-hairs until the error is less than an angular error of 20 seconds.

3.9.6.3 Horizontal axis.- It shall be possible to adjust the horizontal axis so that the telescope will transit in a vertical plane when it is leveled. Horizontal and vertical axes shall be perpendicular.

3.9.6.4 Vertical circle fiducial mark.- The fiducial mark shall be capable of being adjusted so that the error in reading after final adjustment shall be less than 0.01 degree.

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3.9.7 Lighting equipment.- The lighting equipment shall consist of the type and number of lamps as in the model and shall provide sufficient illumination for night operation of the theodolite.

3.9.7.1 Power requirement.- The lighting equipment shall operate from a dc power source such as two Batteries BA-30 conforming to Specification MIL-B-18. The inner dimensions of the battery box shall permit a clearance of not less than 3/64 inch on all sides of each battery. The battery box shall be marked "Use two type BA-30 batteries".

3.9.7.2 Wiring.- All wiring shall be single conductor, 4107 circular mils, No. 18 AWG, stranded wire conforming to Specification MIL-W-76.

3.9.8 Quality of image. The image of both telescopes shall be clear, well defined and free from distortion, coma, and chromatic aberration.

3.9.9 Fiducial surfaces.-

3.9.9.1 Edges.- The edges of these surfaces shall be concentric with the scale edges.

3.9.9.2 Faces.- The face of each of these surfaces shall lie in the same plane as the face of the scale with which it is used.

3.9.9.3 Separation.- There shall be no noticeable separation between the edges of the surfaces and the scales. Any theodolite not meeting these requirements shall be readjusted by the contractor to demonstrate that they can be made to do so before acceptance.

3.9.9.4 Engraving.- The engraving shall be narrow, clean-cut and of uniform width and depth on all scales. It shall be uniformly pigmented and form a sharp, readable contrast with its background. The lines shall be of uniform width to the edges of the scales. The numbering shall be the same size and style on all scales.

3.10 Case CY-787( )/U.- This case shall be constructed in accordance with Drawing and Data List SC-DL-9073.

3.11 Interchangeability.- Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed upon any unspecified dimension, rating, characteristic, etc.

3.12 Service conditions.- The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement:

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3.12.1 Temperature (see 4.11).-

(a) Operating: Ambient temperature in the range of +125° F to -40° F. (The 150° F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160° F to -80° F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.12.2 Relative humidity (see 4.10).- Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours over the number of cycles specified in 4.10. (See 3.6.2)

3.12.3 Elevation (see 4.8).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.13 Bounce.- The theodolite, mounted in its carrying case, shall meet the requirements of 3.8.1 and 3.8.2 and shall suffer no physical damage. Only minor surface abrasion of the case is acceptable.

3.14 Marking.General.- Marking shall conform to Specification MIL-M-13231. (See 4.4.)

3.15 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.1.)

3.16 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268:

Wiring and cabling, including:
- Slack.
- Protection.
- Clearance.
- Splicing.
- Connections, general.

Flux and cleaning agents for soldering.

Process for soldering.
Cleaning.
Riveting.
General requirements for securing of parts.
Cleaning of welding.
Process for welding.
Mechanical operation of controls.
Gears.

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(b) In this specification:

3.6.1 Assembly	3.8.1 Needle balance
3.6.2 Prism adjustment	3.8.3 Adjustment
3.6.3 Focusing	3.9.1 Levels
3.6.4 Eye piece	3.9.2 Scale concentricity
3.6.6 Lenses	3.9.5.1 - 3.9.5.2 Adjustment
	3.9.9) Fiducial surfaces

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)

(2) Quality conformance inspection of preparation for delivery. (See 4.26.)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A and group B, (see table II and 4.5.2, respectively). The preproduction inspection will be performed in any order satisfactory to the Government.

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Table I.- Preproduction inspection.

Inspection (For additional preproduction inspection see 4.3)	Req Para.	Insp Para.
Bounce	3.13	4.7
Moisture-resistance	3.12.2	4.10
Temperature	3.12.1	4.11
Elevation	3.12.3	4.8

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.14
Lenses	3.6.6

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government-- at its discretion-- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.5.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

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Table II.- Group A inspection.

Inspection	Req Para.	Insp Para.	AQL Major	AQL Minor
Visual and Mechanical	3.16	4.6	1.0%	4.0%
Theodolite main telescope	3.6			
Finder Telescope ML-146( )	3.7			
Compass ML-197( )	3.8			
Case CY-787( )/U	3.10	4.6		
Operational				
Focus, collimation	3.6.3 thru 3.9.6.2.1	4.12		*
Adjustment of vertical axis	3.9.6.1	4.13	1.0% for the group	
Adjustment, horizontal axis	3.9.6.3	4.14		
Adjustment, vertical circle fiducial	3.9.6.4	4.15		
Level sensitivity	3.9.1	4.16		
Tangent mechanism	3.9.5	4.17		
Looseness of centers	3.9.3	4.18		
Obliquity	3.9.4	4.19		
Optical characteristics	3.6.7 3.7	4.20		
Quality of image	3.9.8	4.21		
Mirror lever	3.7.1	4.23		
Compass	3.8	4.24		
Lighting equipment	3.9.7	4.25		*
Sight accuracy	3.6.5	4.22		

* All operational defects are in major category

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to 4.9 and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.6 Visual and mechanical inspection.- The equipment shall be inspected for conformance to 3.16. Major and minor defects shall be as defined in Standard MIL-STD-105.

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4.7 Bounce test.- The equipment shall be tested on the package tester, type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(b) Place the equipment in its transit case, CY-787( )/U, prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ± 2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.8 Elevation test.- The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.12 thru 4.18	Adjustment

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.8(a). The equipment shall meet specified performance for those measurements.

(c) Further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. After stabilization, repeat the measurements specified in 4.8(a). The equipment shall meet specified performance for these measurements.

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4.9 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3. 11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.10 Moisture-resistance test. - The equipment shall be tested as follows:

4.10.1 Test conditions. -

(a) Do not move equipment from the humidity chamber for measurements.

(b) Complete measurements as rapidly as possible.

4.10.2 Test method. -

(a) Dry at 130° ± 5° F for 24 hours.

(b) Condition at 77° ± 5° F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.10.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.10.3 shall be made during the periods shown on the standard and shall comply with 4.10.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.10.3.

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4.10.3 Performance.- Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
4.12 thru 4.18	Adjustment	3.9.6

4.10.4 Failure.- If any equipment fails to meet the performance specified in 4.10.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.10.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.10.2(e), it does not pass the test.

4.11 Temperature test.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Performance shall be as follows:

Step	Paragraph	Measurement	Performance
1, 4, 8 and 10	4.12 thru 4.18	Adjustment	(1) See 3.9.6 (2) There shall be no bending of moving parts. (3) No change in looseness of fit between steps 1 and 8 (See 3.9.3)

4.12 Focus, collimation.- Before adjustments of collimation are made, the angle of intersection and the alignment of the cross-hairs in the field of the telescope shall be checked. If the intersection of the cross-hairs is not closely centered in the field (as determined by visual observation), it shall be centered by adjusting the screws holding the cross-hair reticule. The alignment of the line of collimation with the horizontal axis shall be checked by the following methods. Focus the main telescope on a target approximately 300 feet distant and so positioned that the line of sight is approximately horizontal. Note the position of the cross-hair intersection on the image of the target. Transit the telescope (rotate it 180 degrees through the zenith about its horizontal axis.) Note the position of the cross-hair intersection on the image of a second target placed approximately the same distance from the telescope as the first target. Rotate the telescope about its vertical axis until the point previously sighted on the first target is again on the cross-hair intersection. Again transit the telescope and sight on the second target. The horizontal distance between the first and second sights on this target is four times the error of collimation. This error may be corrected by adjusting the cross-hair reticule horizontally so that the cross-hair intersection is moved across the image of the target a distance equal to one-fourth the horizontal distance between the first and second sights on the target. The direction of movement of the cross-hairs should be from the second toward the first sight. It will probably be necessary to repeat this adjustment several times before a satisfactorily small error is obtained.

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4.12 Focus, collimation.- (Cont'd)

If the collimation error is sufficiently large to require moving the cross-hair intersection a noticeable distance from the center of the field, this error shall first be reduced by adjusting the position of the telescope prism until the telescope is in approximate collimation. Final adjustments to accurate collimation shall then be made by the above method. The inspector shall determine that adequate means for making this adjustment have been provided. Where special collimating equipment which will give effectively the same accuracy as a 300 feet distant target is available, it may be substituted for the above cited target.

4.12.1 Change in collimation of the telescope with change in focus.- The collimation of the telescope shall be checked by the method given in 4.12 when focused on a point sufficiently distant to be equivalen to infinite focus, and any error of collimation noted. The telescope shall be refocused on a target at minimum focal distance from the telescope and the error of collimation redetermined.

4.13 Adjustment of the vertical axis.- Adjustment of the vertical axis is made by leveling the instrument. In order to accomplish this, the alignment of each level vial with the vertical axis shall be checked. In order to verify the adjustment of the levels, disengage the tangent mechanism of the horizontal circle and turn the instrument so as to place one of the levels parallel to the line joining two of the leveling screws. Center the bubble in the level vial. Rotate the instrument 180 degrees. If the bubble returns between its marks, the level is adjusted. If not, correct half of the difference by means of the adjustment provided. Repeat this operation until the bubble does not vary when the instrument is rotated 180 degrees. The adjustment of each level shall be verified separately as above. Avoid handling or breathing on the levels during adjustment.

4.14 Adjustment of the horizontal axis.- The following method shall be used in checking this adjustment. The instrument shall be leveled carefully and focused with the intersection of the cross-hairs on the top of a vertical plumb line or building corner. With the vertical axis stationary, the telescope shall be revolved about its horizontal axis until it is focused at the bottom of the line or corner. The telescope shall then be revolved about its vertical axis and transited until it is again focused on the same point at the top of the line. It shall again be revolved about its horizontal axis to the bottom of the line. Any difference between the two positions of the cross-hairs at the bottom point represents twice the error alignment between horizontal and vertical axes. The adjustable bearing of the horizontal axis shall be adjusted for one-half the difference between the lower sightings. It shall be possible to adjust the instrument so that the error of alignment is less than an angular error of 20 seconds. In making this test, a target shall be chosen, if possible, which shall permit a rotation about the horizontal axis of not less than 45 degrees.

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4.15 Adjustment of the vertical circle fiducial mark. - The method of checking the adjustment of the fiducial mark of the vertical circle shall be as follows: The instrument shall be leveled and the zero of the mark brought into exact coincidence with the zero of the vertical circle. A magnifying glass shall be used to make this adjustment. The position of the cross-hair intersection shall be noted on the image of a target set up for the purpose. The instrument shall be turned through exactly 180 degrees on its horizontal axis, and then through 180 degrees on its vertical axis so that it sights again in the same direction. If the point previously sighted on the target falls on the cross-hair intersection, the zero of the vertical circle is properly adjusted. If not, the telescope shall be moved to obtain this result and the difference in reading of the vertical circle graduated drum noted. The position of the vertical circle fiducial mark shall be corrected by one half of the difference of reading, by the adjusting means provided. See 3.9.6.4.

4.16 Level sensitivity. - The level sensitivity and hysteresis shall be checked at or near the center of each level vial. See 3.9.1.

4.17 Tangent mechanisms. -

4.17.1 Mesh engagement. - The tangent mechanisms shall be placed in the in-mesh position and the telescope rotated as rapidly as possible about each axis by means of each tangent mechanism in turn. Neither tangent mechanism shall exhibit a tendency to slide out of engagement.

4.17.2 Tangent screws. - The tangent mechanisms on each theodolite shall be checked to determine that the tangent screws are accurately aligned with the worm gears on the vertical circle and intermediate center plate, respectively. To make this test, each tangent mechanism shall be placed in the in-mesh position, and then carefully removed. The alignment between the surfaces on which the scales and their respective fiducial marks are engrave shall not change in any way as the tangent mechanisms are released. A magnifying glass shall be used to observe the scales.

4.17.3 Backlash. - The tangent mechanisms shall be tested for backlash in the in-mesh position.

4.17.4 Gear accuracy. - The accuracy of the gears on each circle and the worm thread on tangent screw shall be tested. The method of test shall be as follows: A degree mark on the scale shall be very carefully aligned with its fiducial marking using a magnifying glass or similar means for scale magnification. The graduated drum shall be adjusted for zero reading. The circle shall be rotated one degree and again carefully aligned with the fiducial mark. The error in the graduated drum indication shall be noted. The error shall be determined from the average of at least three readings made in the same direction of rotation on the same degree mark. This procedure shall be repeated for at least six points around the scale circumference. The error for each point shall not exceed 0.1 degree. The scale shall be carefully rotated approximately 90 degrees. The fiducial mark shall be carefully aligned at the beginning and end of this traverse. The total cummulative error on the graduated drum shall not exceed .02 degrees.

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4.17.5 Azimuth and vertical settings. - A target in sharp focus shall be accurately sighted on the cross-hairs, employing the finder telescope. The azimuth and vertical settings shall be carefully noted. The swining mirror shall be moved so that the finder telescope shall refocus on the target five times. At each of these, refocusing readings shall be taken. The image of the target shall not appear to have shifted more than 40 seconds between any two consecutive readings for either scale.

4.18 Looseness of centers. - The theodolite shall be placed on a metal plate or similar unyielding surface, carefully leveled. Both tangent mechanisms shall be placed in the out-of-mesh position, and the intermediate-center clamp released. The telescope shall be focused on a scale with the line of sight approximately horizontal. A weight of approximately two pounds shall be placed in turn on opposite sides of the theodolite base-plate on a line parallel to the objective end of the telescope, and rotated approximately 90 degrees and the base-plate approximately 180 degrees, and the test repeated. The total movement of the cross-hairs with the weight on opposite sides of the plate shall determine looseness of the centers.

4.19 Obliquity of the intermediate-center axes. - The theodolite shall be rigidly supported and leveled. The tangent mechanisms shall be placed in the out-of-mesh position and the intermediate-circle clamp released. With the telescope and base-plate maintained in a stationary position, the intermediate center shall be turned through a complete revolution. The telescope shall be focused on a scale and the total shift of the cross-hairs measured. The error thus determined is twice the error of obliquity.

4.20 Optical characteristics. - The optical characteristics of the telescopes shall be measured for conformance with the requirements of 3.6.7 and 3.7.

4.20.1 Test procedure. - The contractor shall submit to the contracting officer for approval, a complete detailed description of the inspection and test procedure which he proposed to use to insure compliance with 4.20.

4.21 Quality of image. - Both telescopes of each theodolite shall be focused in turn on a number of objects located at various distances beyond the minimum focusing distance from the theodolite to determine compliance with 3.9.8.

4.22 Accuracy of sights. - The theodolite shall be accurately sighted on a fixed object by means of the pair of sights located close and parallel to the telescope mounting. Without moving the telescope it shall be focused on this object. The image of the object sighted shall lie not more than one-fourth the field diameter from the intersection of the cross-hairs. This test shall be repeated using the folding pair of sights of the theodolite in the raised position to determine compliance with 3.6.5.

4.23 Mirror lever adjustment. - Each theodolite shall be found to comply with the following test. The telescope assembly shall be rotated from below the horizontal to an elevation of 90 degrees with the swinging mirror so placed that the image can be viewed through the finder telescope. The mirror lever and mirror shall be found to retain their given position at any angle of elevation. The mirror shall be swung to in and out positions and the theodolite tested for compliance with 3.7.3 and 3.6.4.

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4.24 Compass.-

4.24.1 Pole visibility.- Each compass shall be tested to ascertain that the south-seeking pole is visible through the compass window.

4.24.2 Accuracy.- The compass needle shall be loosened and the theodolite oriented on magnetic north, as indicated by the compass needle. The position of the instrument on a target shall be noted. The compass shall then be locked, and the instrument rotated through approximately 360 degrees. The compass needle shall again be loosened, and the cross-hairs of the instrument lined up with magnetic north as indicated by the needle. The second reading on the target shall not differ from the first by more than one degree. Care shall be taken to keep the mount of local attraction constant during test.

4.24.3 Magnetism of the needle.- This test for magnetism shall be made on appratus consisting of a compass with 1-1/2 inch needle, mounted on a board which has a groove 1/10 inch wide in the center to receive the compass needle which is to be tested. The zeros of the compass shall be in a plane with the center of the groove in the board. A location with minimum magnetic attraction shall be selected and the board placed in such a position that when the needle in the compass is released to swing on its pivot, it will read zero on the compass ring. Then the south end of the compass needle, which is to be tested, shall be placed two inches from the center of the compass. It should then deflect the needle in the testing compass at least 10 degrees. The same test shall be repeated a few days later, to see if there has been any los of magnetism which will be indicated by the smaller deflection of the needle in the compass. This difference in deflection shall not be greater than 5 percent of the original reading.

4.25 Test of lighting equipment.-

4.25.1 Continuity test.- Continuity test shall be made on the lighting circuit with the switches in each combination of operating positions.

4.25.2 Illumination.- The theodolite shall be tested under conditions simulating night observation for compliance with illumination requirements of 3.9.7.

4.26 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.-

5.1.1 Level A.- Each Theodolite, Double Center, ML-474( )/GM, shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116, and described in the following paragraphs:

5.1.1.1 Technical Manual.- Each Technical Manual shall be packaged Method IC-6 as follows: Inclose each Technical Manual within a close-fitting, Type II, Class d bag conforming to Specification MIL-B-117. Heat seal to effect closure.

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5.1.1.2 Accessories and Spare Parts in Case CY-787/U.- No packaging required. Accessories and spare part items shall be placed and secured in the provided spaces within the case.

5.1.1.3 Theodolite, Double Center ML-474( )/GM in Case CY-787/U.- Package each Theodolite Method IIb as follows: Secure the Theodolite to the mounting base of the case. Close case and secure fastening. Cushion each cased Theodolite on all surfaces with cells or pads or both, fabricated of corrugated fiberboard, Type CF, Class Domestic, Variety SW, Grade 200, conforming to Specification PPP-F-320 and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned Theodolite together with the required number of units of activated desiccant within a close-fitting, corrugated fiberboard box, Type I, Class 1, Style RSC conforming to Specification PPP-B-636. The desiccant conforming to Specification MIL-D-3464 shall be placed within one of the cushioning cells in the quantity determined by Formula 1 of Specification MIL-P-116. A humidity indicator card, conforming to Military Standard MS-20003, shall be secured to the underside of one of the top inner flaps of the container immediately inclosing the contents. The indicator card shall be secured with water-resistant, pressure-sensitive tape conforming to PPP-T-76. Closure of the box shall be effected in accordance with the box specification or the appendix thereto. Blunt all corners of the box. Place the boxed Theodolite within a water-vaporproof barrier, Class 1, conforming to Specification MIL-B-131. The barrier shall be fabricated to form a bag with one (1) dimension to allow for three (3) additional reseals if opened for inspection or other purposes. Heat seal the water-vaporproof barrier bag leaving a small opening to exhaust the air. Evacuate the excess air and complete the heat seal. The time lapse between positioning the desiccant and the completion of the barrier bag seal must be held to an absolute minimum. Place the water-vaporproofed Theodolite within a close-fitting fiberboard box, Type I, Class 2, W5c, Style RSC conforming to Specification PPP-B-636. Any stapling on the inside of the exterior box shall be covered with water-resistant, pressure-sensitive tape conforming to Specification PPP-T-76, to prevent rupturing the barrier bag. Closure shall be in accordance with the box specification or the appendix thereto. Secure the Technical Manuals, packaged as specified (See 5.1.1.1) to the top surface of the package container, with water-resistant, pressure-sensitive tape conforming to Specification PPP-T-76.

5.1.2 Level C.- Theodolite, Double Center ML-474( )/GM shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity at the lowest transportation rate.

5.1.3 Package performance.- Package testing shall be performed in accordance with the requirements of Specification MIL-P-116. The rough handling test or cyclic exposure test and functional test will be required only when invoked in the bid request or contract. (See 6.2.)

5.2 Packing.-

5.2.1 Level A.- Each Theodolite, Double Center, ML-474( )/GM shall be packed in a close-fitting Style 4, Class 2, nailed wood box conforming to Specification PPP-B-621 or a Type III, Class 2, Style A, cleated plywood box conforming to Specification PPP-B-601. Fabricate the box to fit the contents snugly. Closure shall be made in accordance with the applicable box specification or the appendix thereto.

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5.2.1.1 Metal strapping.- Metal strapping, Type I, Class b, conforming to Specification QQ-S-781 shall be applied in accordance with the requirements of the box specification.

5.2.2 Level B.- Each Theodolite, Double Center ML-474( )/GM shall be packed for shipment as specified in paragraph 5.2.1 except the nailed wood box shall be Class 1, and the cleated plywood box shall be Type IV, Class 1. Closure shall be in accordance with the applicable box specification or the appendix thereto.

5.2.3 Level C.- Theodolite, Double Center ML-474( )/GM shall be packed in containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Rules or Regulations of the common carrier, as applicable to the mode of transportation.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Military Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(c) The specific paragraphs of Section 5 which are applicable to the particular procurement.

(d) If rough handling or cyclic exposure test and functional test are required, so state.

(e) Technical literature required. (See 3.15.)

(f) Quantity of tools and running spare parts required. (See 3.15.)

6.2 Color.- The color chip furnished by the procuring agency (3.4) will be color chip No. X-24087, and may be obtained upon request to Commanding General, U. S. Army Electronics Materiel Agency, 225 South Eighteenth Street, Philadelphia 3, Pennsylvania, ATTN: SELMA-J4b.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: ML-474W/GM. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.4 Definition of level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the Unites States.

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6.5 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Project No. 6660-A082

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426 APR 62

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

C-9379

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Commanding General
U. S. Army Electronics Command
Attn: AMSEL-TD-S
Fort Monmouth, N. J. 07703

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